PHASE 2 - Backend Development & Configurations – Data Architecture

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1. Phase Overview

Phase 2 focuses on developing the backend structure and configuration required for the **Car Rental Request Automation** project. The main objective of this phase is to create the data architecture, configure business rules, and implement automation logic using **ServiceNow Flow Designer**.

The backend configuration connects the Service Catalog request with car inventory, approval processing, catalog tasks, assignment, and email notifications. This creates a structured and data-driven process for handling car rental requests.

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2. Data Architecture

The data architecture defines how the information required for the car rental process is stored and processed in ServiceNow.

The project uses the **Service Catalog**, request records, approval records, catalog tasks, and a custom **Car Inventory** table to support the complete workflow.

Main Data Components

Component	Purpose
Car Rental Request Catalog Item	Captures the details provided by the requester.
Car Inventory Table	Stores information about available cars and their availability.
Request / RITM Records	Stores submitted catalog requests and tracks request progress.
Approval Records	Maintains the approval decision for the request.
Catalog Task (SC Task)	Stores fulfillment work assigned to the Transport Team.
Flow Designer	Automates the request, approval, task creation, and notification process.

Car Rental Request Variables

The catalog item collects the following information:

- Requester

- Requester Email
- Requested Date
- Pickup Location
- Drop Location
- Duration (Hours)
- Car Type
- Reason

These values are captured when the requester submits the **Car Rental Request**.

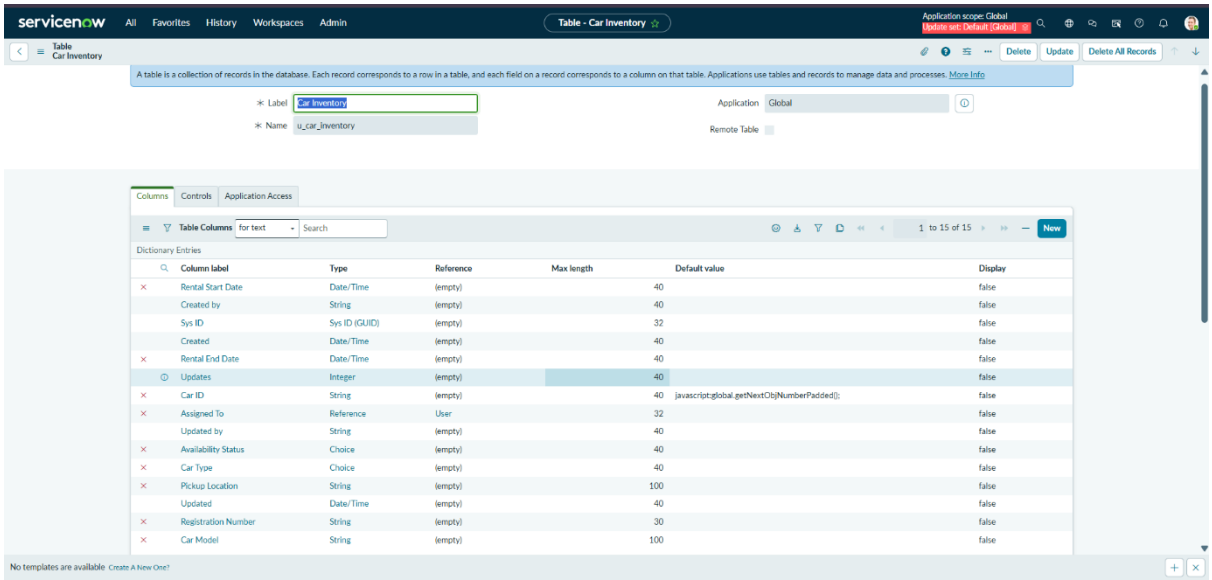
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3. Car Inventory Configuration

A **Car Inventory** table was configured to maintain the information related to the cars available for rental.

The table is used as a supporting data source for the rental request process. It helps the application determine whether a required vehicle is available and supports the car allocation process.

The inventory configuration provides a structured way to maintain vehicle information instead of managing availability manually.



Caption: Car Inventory table configured in ServiceNow.

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4. Business Rules

Business Rules were configured in ServiceNow to implement backend processing and validation.

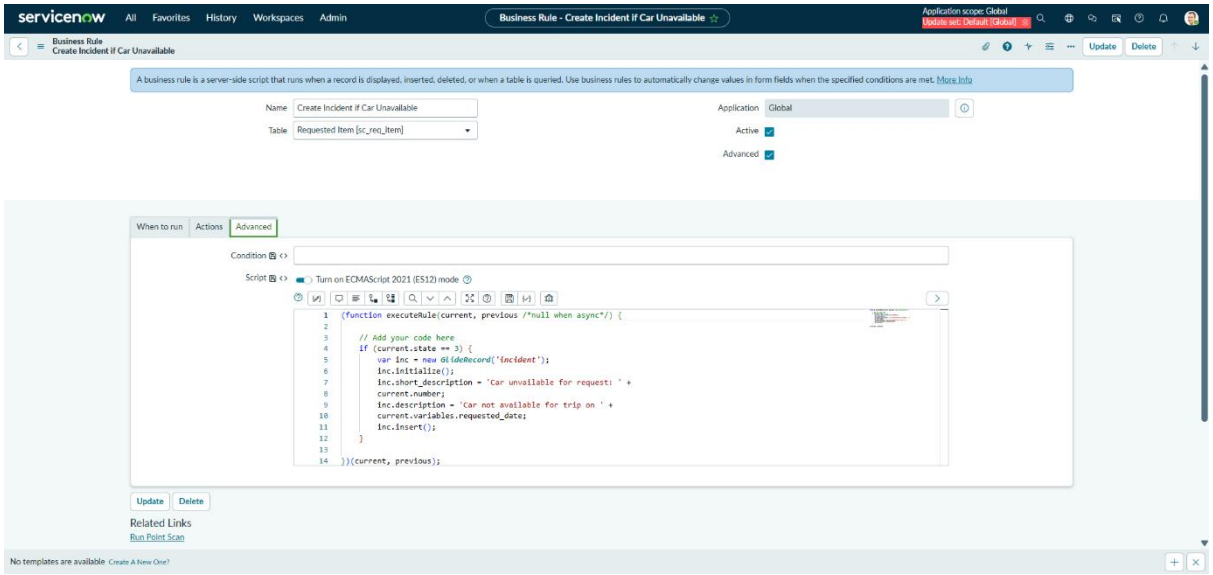
A **Car Unavailable** Business Rule was created as part of the project. The rule supports the validation of car availability and helps identify situations where a requested vehicle cannot be provided.

Business Rules provide server-side automation so that important processing can happen automatically when records are created or updated.

Business Rule Objective

The Business Rule is used to:

- Validate car availability.
- Identify unavailable vehicles.
- Support backend processing of car rental information.
- Reduce the need for manual checking.
- Maintain consistent processing of inventory information.



Caption: Car Unavailable Business Rule configured in ServiceNow.

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5. Automation Logic

The project uses **ServiceNow Flow Designer** to automate the car rental request process.

The workflow is triggered when a **Car Rental Request** is submitted through the Service Catalog. The flow retrieves the information entered by the requester and uses it during the approval and fulfillment process.

Main Flow Steps

Step 1 – Catalog Item Requested

The flow starts when the **Car Rental Request** catalog item is submitted.

Step 2 – Get Catalog Variables

The flow retrieves the information entered in the catalog form, including:

- Requester
- Requested Date
- Pickup Location
- Drop Location
- Duration
- Car Type
- Reason

Step 3 – Ask for Approval

The submitted request is sent through the configured approval process.

The approver reviews the request and provides an approval decision.

Step 4 – Approval Decision

The flow evaluates the approval result.

If the request is approved, the workflow continues with the fulfillment process.

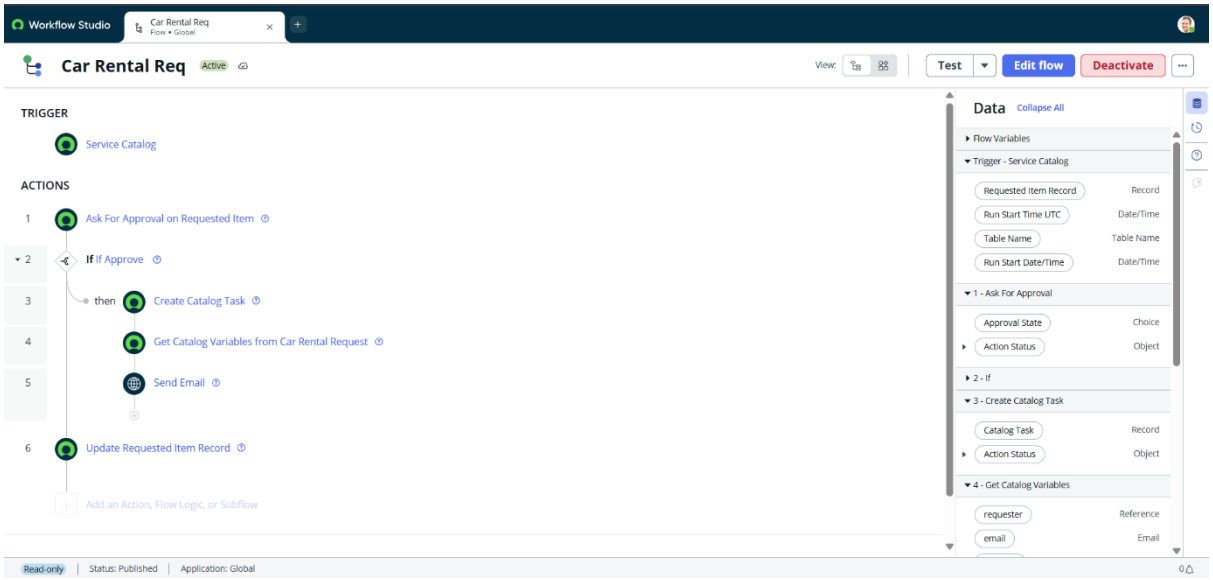
Step 5 – Create Catalog Task

After approval, a **Catalog Task (SC Task)** is created.

The task is assigned to the **Transport Team** for fulfillment.

Step 6 – Send Email Notification

An email notification is sent to the requester to communicate the approval status and request progress.



Caption: Car Rental Request automation flow configured using Flow Designer.

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6. Data-Driven Workflow Approach

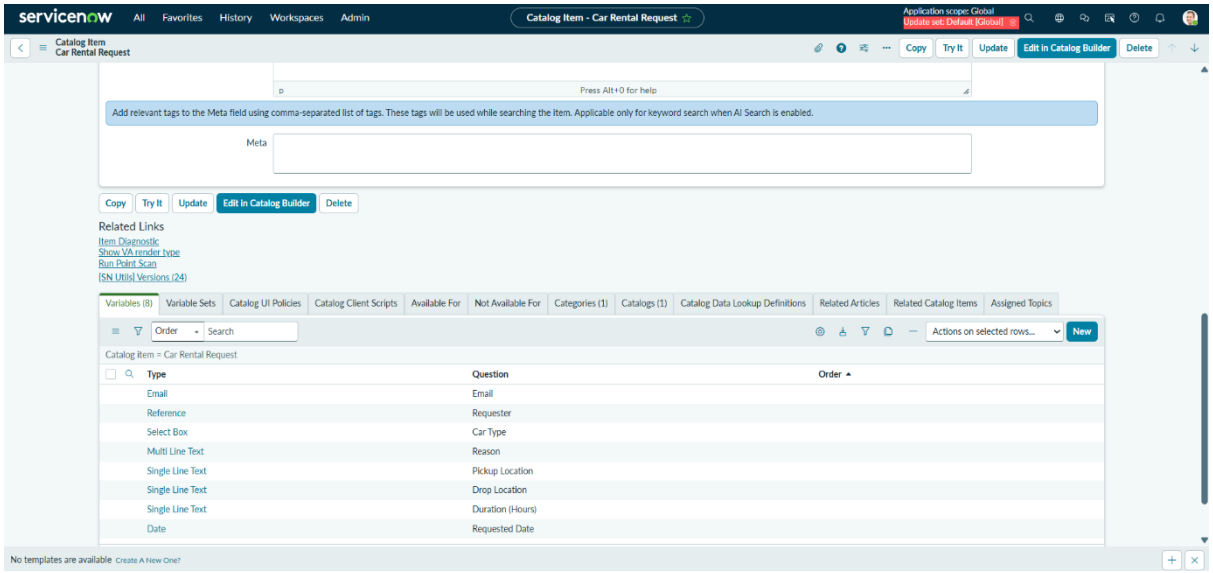
The workflow follows a **data-driven approach**, meaning that the automation uses the information captured from the Service Catalog rather than requiring the same information to be entered manually at every stage.

For example, the selected **Car Type**, rental duration, requested date, requester information, pickup location, and drop location are captured from the catalog request and made available to the workflow.

This allows the workflow to use the submitted request information during approval, task creation, and notification activities.

Catalog Variables Used by the Workflow

Variable	Purpose
Requester	Identifies the person submitting the request.
Requester Email	Used for sending notifications.
Requested Date	Specifies the planned rental date.
Pickup Location	Specifies where the vehicle is required.
Drop Location	Specifies the destination.
Duration (Hours)	Defines how long the vehicle is required.
Car Type	Specifies the requested vehicle category.
Reason	Provides the purpose of the rental.



Caption: Car Rental Request catalog item and configured variables.

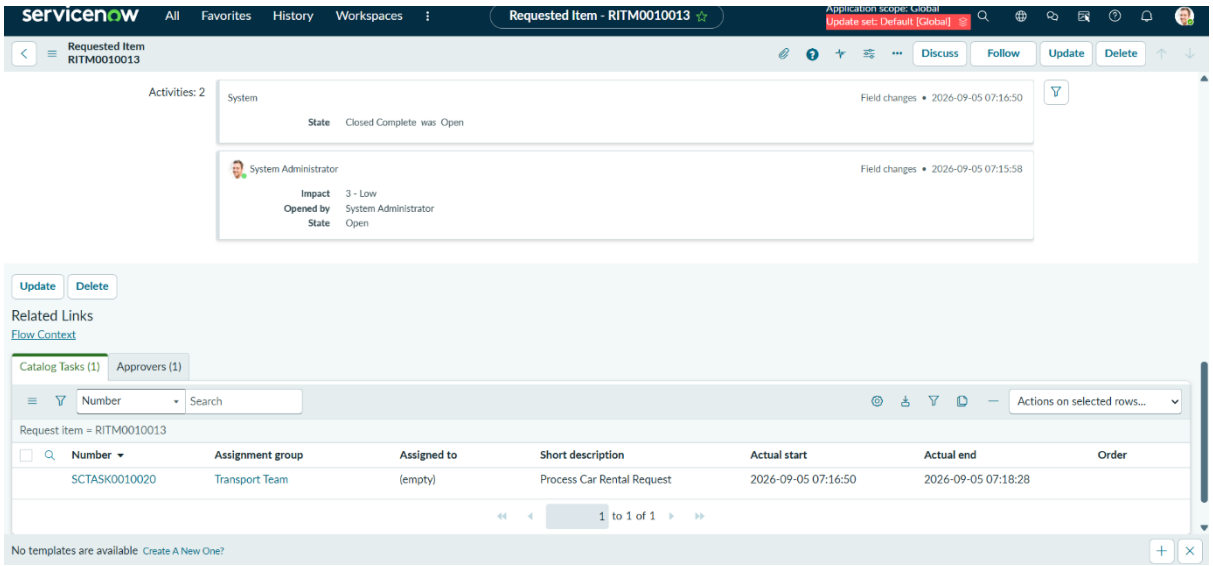
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7. Catalog Task and Assignment

After the request passes the approval stage, a **Catalog Task** is created for fulfillment.

The Catalog Task contains the information required by the fulfillment team and is assigned to the **Transport Team**.

This ensures that approved car rental requests are transferred to the appropriate team for further processing.



Caption: Catalog Task created and assigned to the Transport Team.

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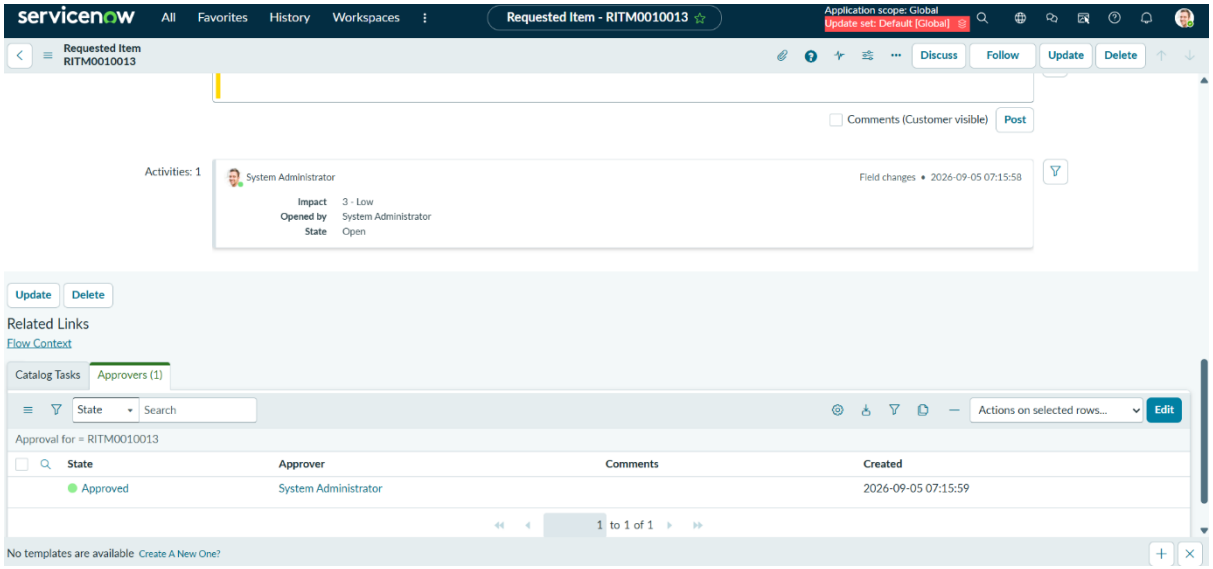
8. Approval and Notification Process

The backend automation also supports the approval and notification process.

The general process is:

Car Rental Request → RITM Created → Approval → Approval Decision → Catalog Task → Transport Team → Email Notification

This provides a structured workflow and allows the progress of a request to be tracked through the ServiceNow platform.



Caption: Request item and approval information in ServiceNow.

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9. Phase 2 Outcome

The backend development and configuration activities establish the foundation required for the Car Rental Request Automation project.

During this phase:

- The required data architecture was defined.
- The Car Inventory table was configured.
- The Car Rental Request catalog item and variables were configured.
- A Business Rule was created for car availability processing.
- Flow Designer automation was configured.
- Approval processing was incorporated into the workflow.
- Catalog Task creation and assignment to the Transport Team were configured.
- Email notification processing was included.

These configurations provide the backend support required for the end-to-end car rental request process.

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10. Phase 2 Conclusion

Phase 2 successfully established the backend configuration and data-driven automation structure for the **Car Rental Request Automation in ServiceNow** project.

The configured architecture connects the Service Catalog, inventory information, approvals, catalog tasks, fulfillment assignment, and notifications into a single automated process. This backend foundation supports the subsequent testing, customization, and final deployment activities of the project.